

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	13 July 2026
Team ID	
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architectural diagram below along with the information in Table-1 and Table-2.

Visualizing Housing Market Trends — Architecture Flow

Guidelines:

- Include all the processes (as an application logic / technology block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third-party APIs etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

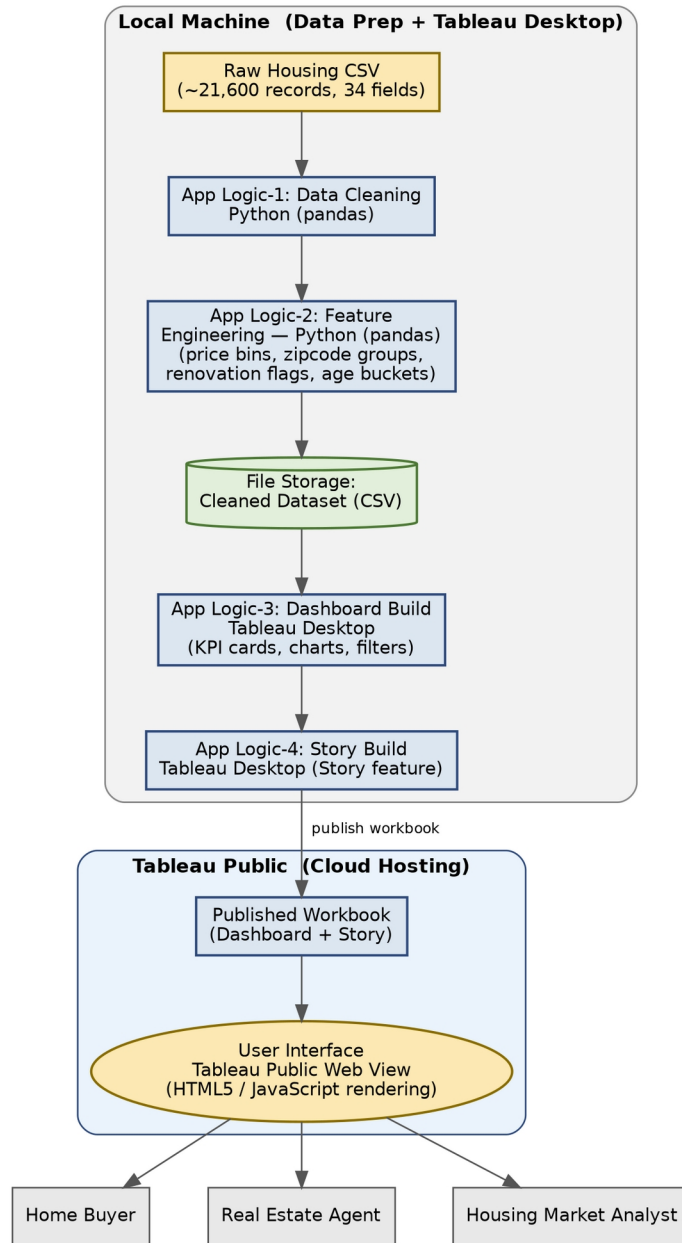


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How users view housing market insights, directly through the published Tableau Public dashboard and story in a web browser	Tableau Public (Web-based Interactive View), HTML5 / JavaScript (Tableau rendering)
2	Application Logic-1	Cleans and prepares the raw housing CSV dataset prior to visualization (handling missing values, formatting fields, etc.)	Python (pandas)
3	Application Logic-2	Transforms prepared data into engineered features used by the dashboard (sale-price bins, zipcode grouping, renovation flags, age buckets)	Python (pandas)
4	Application Logic-3	Builds the interactive KPI cards, charts and filters that make up the housing dashboard	Tableau Desktop
5	Application Logic-4	Builds the guided Tableau Story narrating key housing-market insights	Tableau Desktop (Story feature)
6	Database	This project does not use a traditional database; data is sourced directly from a CSV file	Not Applicable for this project
7	Cloud Database	Not Applicable for this project	Not Applicable for this project
8	File Storage	Storage of the raw and cleaned/transformed housing dataset used to build the dashboard	Local File System (CSV)
9	External API-1	Not Applicable for this project	Not Applicable for this project
10	External API-2	Not Applicable for this project	Not Applicable for this project
11	Machine Learning Model	Not used in the current scope; predictive price modelling was identified as future scope during idea prioritization	Not Applicable for this project
12	Infrastructure (Server / Cloud)	Application Deployment: dashboard and story published directly to Tableau Public (no dedicated server or custom website); local system used for data cleaning and Tableau Desktop workbook development	Local Machine (Data Prep + Tableau Desktop), Tableau Public (Dashboard & Story Hosting)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source Python libraries used for data cleaning and feature preparation	pandas, NumPy
2	Security Implementations	Public Tableau dashboard exposes only aggregated, anonymised property features; no personally identifiable buyer or seller information is published	Tableau Public access settings

3	Scalable Architecture	Additional worksheets, dashboards, or datasets can be added to the Tableau Public workbook without restructuring the overall project	Modular Tableau workbook structure
4	Availability	Dashboard and story remain accessible anytime via the Tableau Public link, with no dependency on a dedicated hosting server	Tableau Public hosting
5	Performance	Optimised visualisations using pre-aggregated fields (sale-price bins, zipcode groups) and filtered extracts for fast load and filter response	Tableau Public rendering optimisation

References

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>